EQUITY IN AUTOIMMUNITY:

Mapping Overlapping Genetic Variants in Underserved Populations

MAKENZIE ROULAE WIGGINS

Final Presentation | Data Wrangling

THE PERSONAL WHY

A Brother's Story

At age 2, my brother suffered a severe anaphylactic reaction.

Three years later
— despite doctors dismissing symptoms — he was diagnosed with Type 1 Diabetes.

This sequence of events — a massive immune event followed by T1D onset — became the central question of this research.

Black individuals with T1D are statistically more likely to experience ketoacidosis, and representation in the research that could help them is critically low.

2

Age at anaphylaxis

years old

3

Years until T1D diagnosis

years later

DKA

Black T1D patients at higher risk of

diabetic ketoacidosis

RESEARCH QUESTION

What shared variants/loci exist between severe early-onset allergic phenotypes and T1D, and are these loci underrepresented in Black populations within the GWAS Catalog and ClinVar?

Type 1 Diabetes

Severe Allergic Phenotypes

Black Population Representation

Three pillars — brought together for the first time

Related Works

Landrum et al. (2014) — Nucleic Acids Research

ClinVar: Public Archive of Relationships Among Sequence Variation and Human Phenotype

Use of the ClinVar variant_summary as a clinical validation layer. This is the foundational paper establishing ClinVar as the standard bridge between raw genetic variants and clinical phenotypes.

Balding et al. (2006) — Current Protocols in Human Genetics

Analyzing Human Genetic Variation Using SNP-Based Methods (doi: 10.1002/0471142905.hg0816s89)

Establishes the methodological precedent for using SNPs to identify disease-associated loci. This is the "why SNPs" justification for filtering and loci-identification approach.

Ng et al. (2014) — PLOS Genetics

Meta-Analysis of Genome-Wide Association Studies in African Americans Provides Insights into the Genetic Architecture of Type 2 Diabetes

Provides the methodological blueprint for filtering GWAS data specifically to Black/African American participants. Justifies the 70% Black ancestry threshold filter and the decision to analyze this population separately.

Pomales-Matos et al. (2025) — Frontiers in Genetics

Ancestry Gaps in Cardiovascular GWAS: A Multi-Database Review of African Representation in Genomic Studies

Documents the systemic underrepresentation of African ancestry populations across GWAS databases — providing direct support that shared T1D–allergy loci have never been meaningfully studied in Black populations.

THE DATA

ClinVar / NCBI

variant_summary.txt

8,887,816 rows

43 columns

0 null values

Industry-standard bridge between raw genetic variants and clinical phenotypes. Used to identify T1D and anaphylaxis-related variants and validate shared SNPs.

GWAS Catalog

Associations data

1,061,816 rows

38 columns

5,800+ unique diseases

Population-scale genotype-phenotype associations. Used to find shared SNPs across T1D and allergic phenotypes at scale.

GWAS Ancestry

Ancestries data

39,980 rows

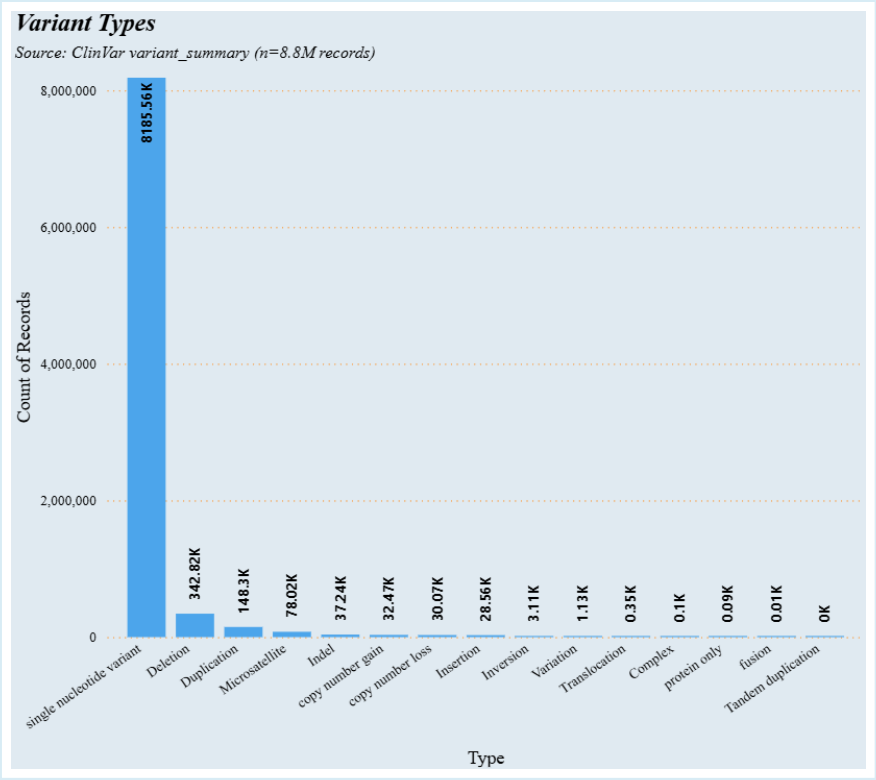
17 columns

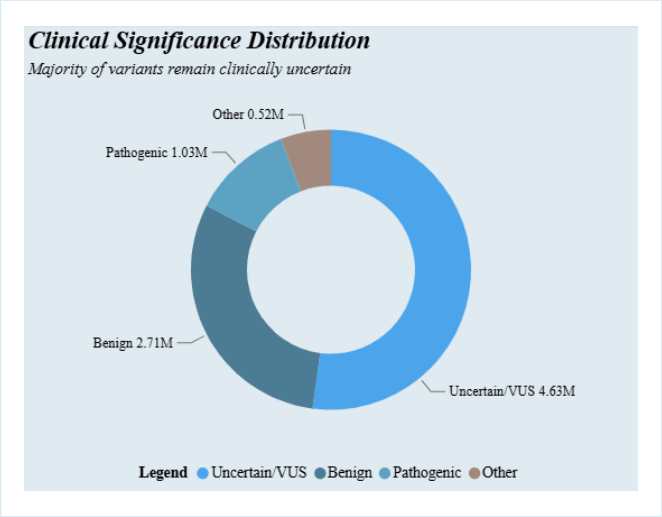
70%+ Black filter

Demographic data for each GWAS study. Used to isolate studies with predominantly Black participant populations.

CLINVAR ANALYSIS — VARIANT TYPES & CLINICAL SIGNIFICANCE

Source: ClinVar variant_summary | 8,887,816 rows | 43 columns | 0 null values

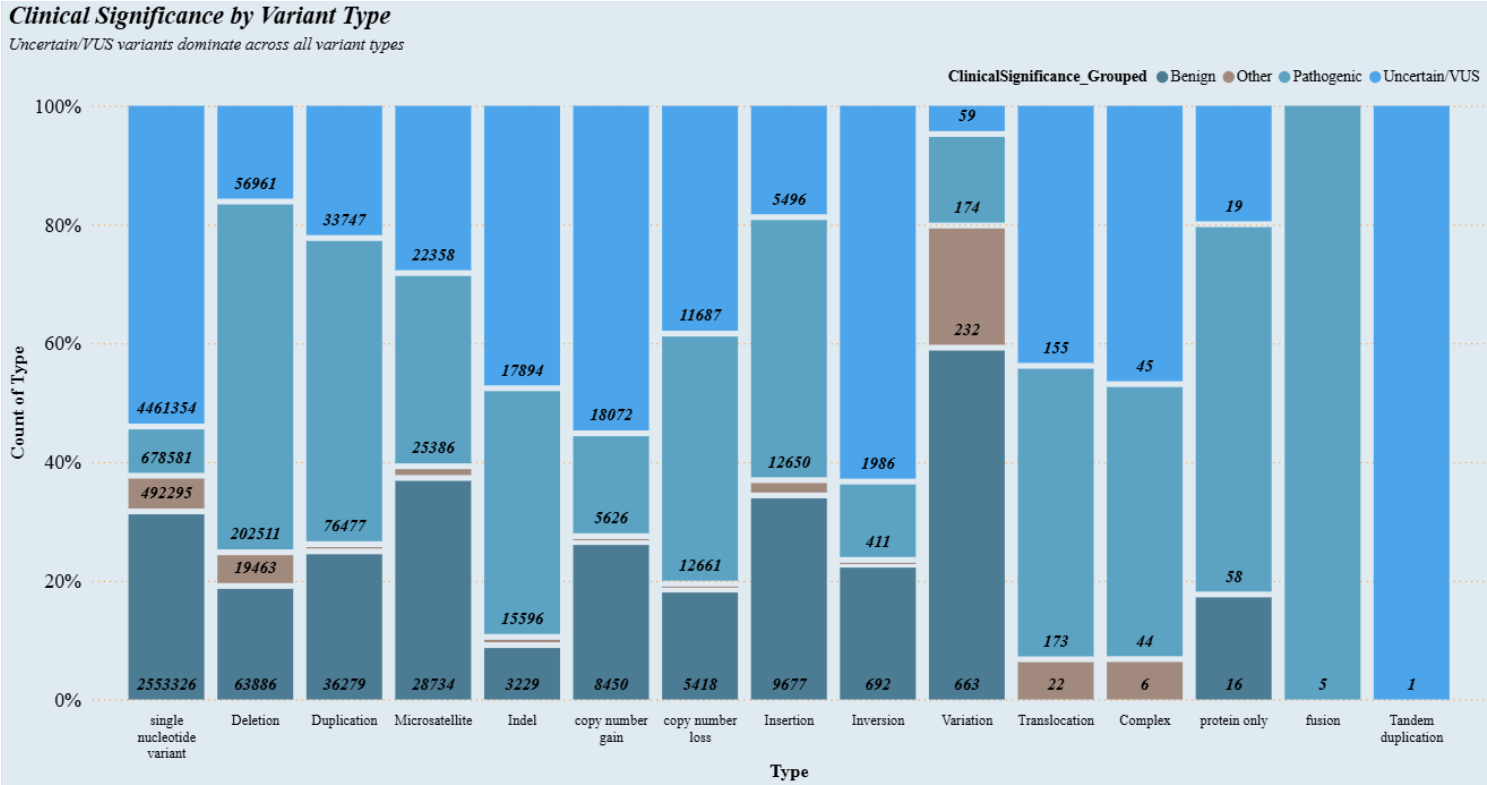




5.8K

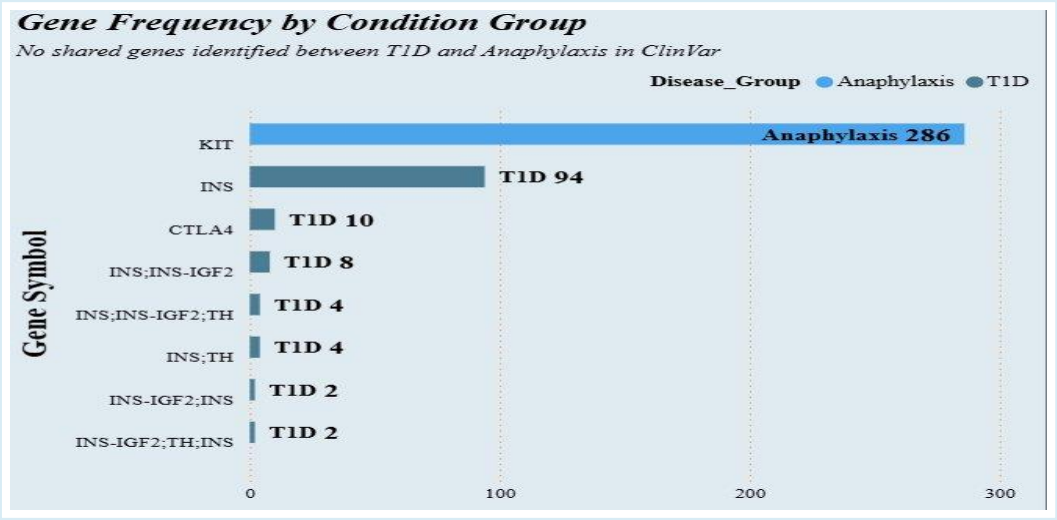
Unique Diseases in GWAS Catalog

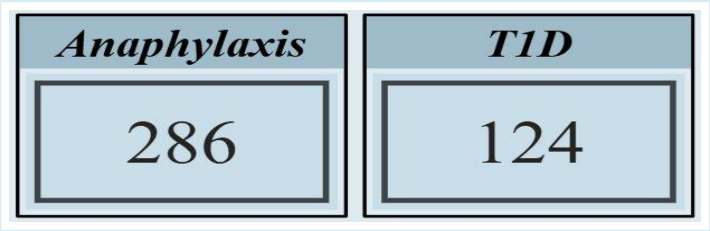
CLINVAR ANALYSIS — CLINICAL SIGNIFICANCE BY VARIANT TYPE

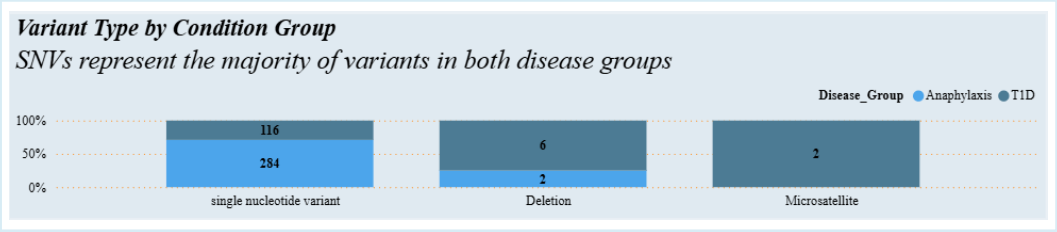


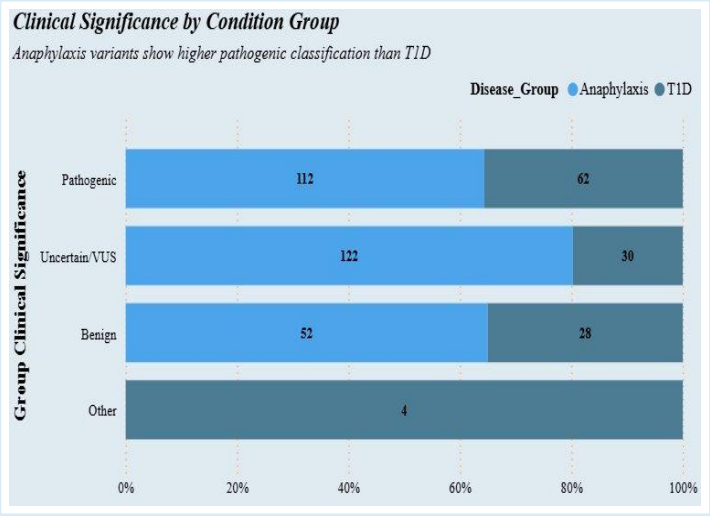
T1D vs ANAPHYLAXIS — GENE OVERLAP IN CLINVAR

No shared genes identified between T1D and Anaphylaxis in ClinVar — supporting the representation gap hypothesis



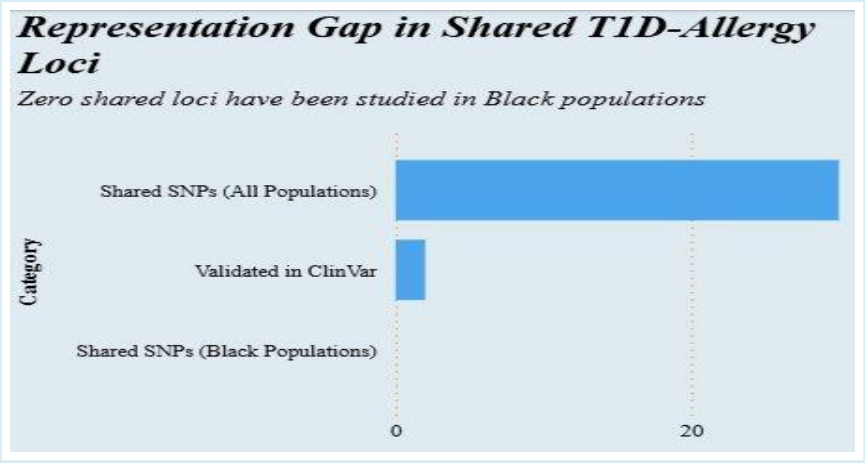


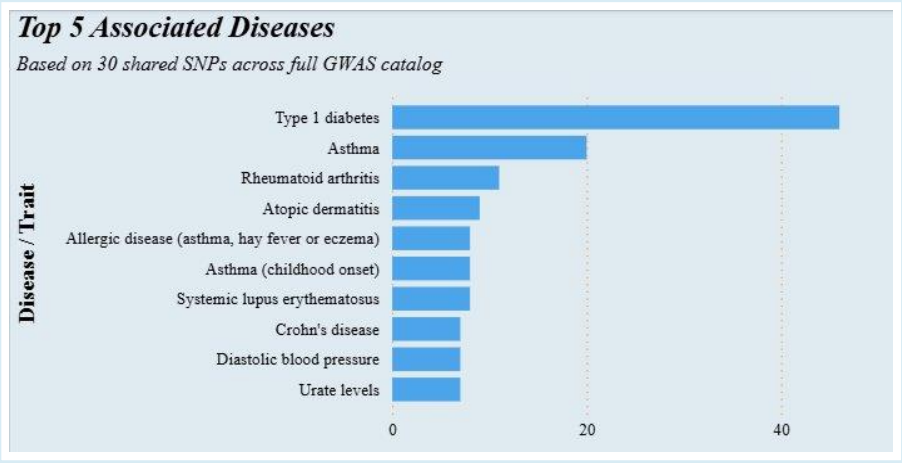


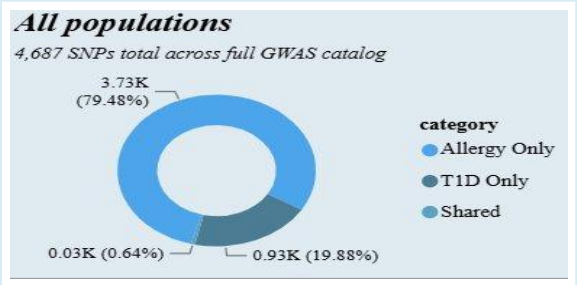


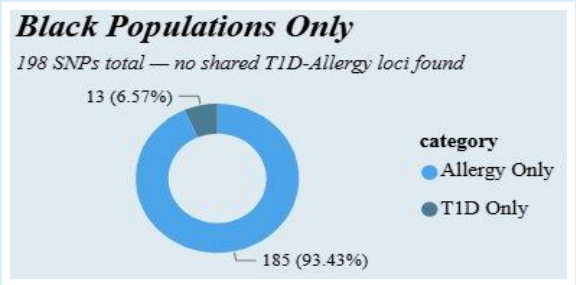
GWAS CATALOG — REPRESENTATION GAP ANALYSIS

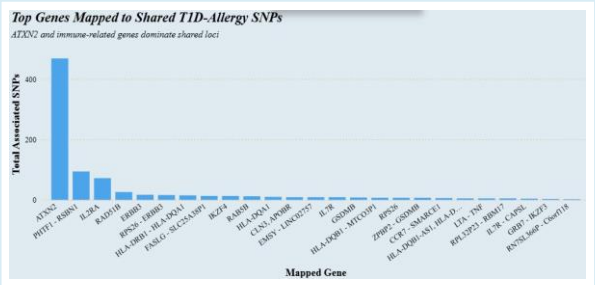
Comparing shared T1D-Allergy SNPs across full GWAS, Black-ancestry filtered studies, and ClinVar clinical validation











WHERE I GOT STUCK

Filtering Black-ancestry GWAS data + joining with T1D & allergy subsets → **0 shared SNPs**

?

Scientific?

Is there truly no genetic overlap between these conditions in Black populations?

?

Filters too strict?

Did my rigid string matching on disease terms miss semantically similar phenotypes?

?

Representation gap?

Have these shared loci simply never been studied in Black populations at scale?

The answer turned out to be all three — and LLMs helped untangle them.

HOW LLMs GOT ME UNSTUCK

Gemini 2.5 Flash integrated directly into the R environment via the GenAI package

01

Feature Type Inference

AutoDW (Liu et al., 2024)

Sampled 3 values per column from the 43-column variant_summary and asked Gemini to classify each as Numerical, Categorical, Datetime, etc. and flag columns influential for T1D or allergy analysis.

02

Demographic Filtering

LLM-Assisted Wrangling

Used Gemini to generate and refine the 70% Black ancestry threshold filter — replacing a brittle 12-condition OR filter with a precise percentage-based approach across study populations.

03

Semantic Phenotype Matching

Zhou et al., 2025

Asked Gemini to remove redundant columns post-join and recommend join strategies when RSID matching failed — bridging ClinVar and GWAS datasets that rigid string filters could not connect.

LLM IN ACTION — SELECTED CODE

Feature Type Inference

```
infer_variant_column_types <-
  function(Variant_data, genai_obj) {

  col_samples <- sapply(
    names(Variant_data), function(col) {
    vals <- na.omit(Variant_data[[col]])[1:3]
    paste0(col,": [",
      paste(vals, collapse=", "),"]")
  })

  prompt <- paste0(
    "Classify each column as: Numerical,",
    " Categorical, Datetime...\n",
    "Flag 'influential' = TRUE if related",
    " to T1D or Allergy phenotypes.\n",
    "Return only raw JSON.\n\n",
    paste(col_samples, collapse="\n"))

  result <- txt(genai.object=genai_obj,
                prompt=prompt)
  fromJSON(gsub("```json|```","",result))
}
```

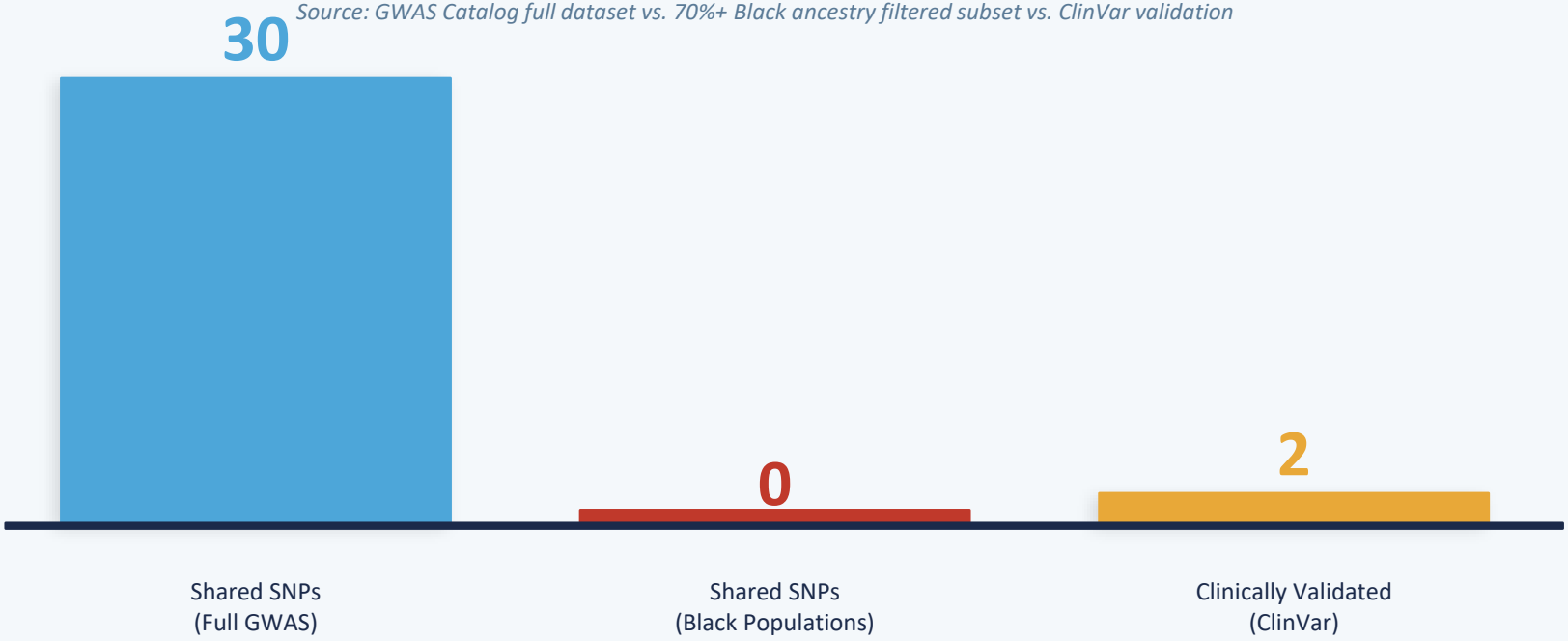
70% Black Ancestry Filter

```
gwas_ancestor_pct <- gwas_ancestor %>%
  mutate(group_n = as.numeric(
    str_replace_all(
      `NUMBER OF INDIVIDUALS`,",",""))) %>%
  group_by(`STUDY ACCESSION`) %>%
  mutate(total_n = sum(group_n,
                       na.rm=TRUE)) %>%
  ungroup() %>%
  mutate(
    is_african = str_detect(
      `BROAD ANCESTRAL CATEGORY`,
      "African American|Sub-Saharan"),
    african_n = ifelse(
      is_african, group_n, 0)) %>%
  group_by(`STUDY ACCESSION`) %>%
  mutate(study_pct_african =
    sum(african_n, na.rm=TRUE) /
    max(total_n, na.rm=TRUE)) %>%
  ungroup() %>%
  filter(study_pct_african >= 0.70)
```

THE KEY FINDING

Representation Gap in Shared T1D–Allergy Loci

Source: GWAS Catalog full dataset vs. 70%+ Black ancestry filtered subset vs. ClinVar validation



Shared genetic loci between T1D and allergic phenotypes exist — but have never been meaningfully studied in Black populations.

THE 2 CLINICALLY VALIDATED SNPs

Of 30 shared SNPs identified in the full GWAS catalog, only 2 have clinical entries in ClinVar

rs6897932 IL7R

Interleukin-7 Receptor

Associated with:

- Type 1 Diabetes
- Asthma
- Multiple sclerosis
- Systemic lupus erythematosus

IL7R variants affect T cell development and survival — directly relevant to T1D autoimmune destruction of pancreatic beta cells AND allergic inflammation. A shared SNP here makes strong biological sense.

rs705704 RPS26

Ribosomal Protein S26

Associated with:

- Type 1 Diabetes
- Asthma
- Body mass index
- Metabolic syndrome

RPS26 sits in chromosome 12q13 and has been associated with T1D through effects on INS (insulin gene) expression. Its appearance as a shared locus with allergy phenotypes is a novel finding worth highlighting.

FUTURE INTEREST

Short Term

- Expand semantic matching using LLMs to capture allergy phenotype variants missed by rigid string filters
- Integrate gnomAD African ancestry population frequency data for allele frequency comparison
- Narrow focus to specific allergens (e.g., tree nuts) where literature suggests stronger T1D correlation

Mid Term

- Apply ML clustering to identify shared functional pathways between allergy and T1D variants
- Build a classifier to flag variants likely underclassified in Black populations due to representation gaps
- Validate IL7R and RPS26 findings against additional Black-ancestry cohorts

Long Term

- Advocate for mandatory ancestry diversity reporting in GWAS publications
- Develop a public-facing tool for clinicians to identify ancestry-specific risk variants
- Publish findings to contribute to equitable genomic medicine for Black adolescents with T1D

THANK YOU

MAKENZIE ROULAE WIGGINS

- 30 shared T1D-Allergy SNPs exist in GWAS — 0 studied in Black populations
- IL7R & RPS26 are clinically validated shared loci with strong biological relevance
- LLM-assisted wrangling is a legitimate methodological contribution to bioinformatics